

has to arbitrate among 4k new classes perhaps need lot more gradient updates than learnable by meta-learners.

| Name                     | 1 shot       | 2 shot       | 3 shot       |
|--------------------------|--------------|--------------|--------------|
| Kaiser (2-cell/label)    | 48.2%        | 58.0%        | 60.3%        |
| Our model (2-cell/label) | <b>52.6%</b> | <b>62.5%</b> | <b>64.1%</b> |
| Kaiser (3-cell/label)    | 49.7%        | 60.1%        | 63.8%        |
| Our model (3-cell/label) | <b>52.8%</b> | <b>63.0%</b> | <b>67.0%</b> |
| Kaiser (5-cell/label)    | 52.7%        | 63.0%        | 66.3%        |
| Our model (5-cell/label) | <b>54.3%</b> | <b>63.2%</b> | <b>66.9%</b> |
| MAML                     | 44.2%        | 46.5%        | 47.3%        |

Table 4: Test accuracies for 4k way few shot learning

## Language Modeling

Language modeling is the task of learning the probability distribution of words over texts. When framed as modeling the probability distribution of words conditioning on previous text, this is just another online sequence prediction task. The natural dependence on history in this task provides for another use-case of memory. Memory based models have been shown to get improvements over standard RNN based language models (?; ?; ?). In the same spirit, we apply LMNs to this task by taking the PCN as a RNN.

We compare our model directly with the recently published neural cache model of (?) and pointer LSTM of (?). These showcase variant uses of memory to improve the prediction of words that repeat in long text. The baseline model (and PCN) is an LSTM with same parameters as in (?). We compared on common language datasets Wikitext2 and Text8 with memory sizes 100 and 2000 as used in the previously published work. We used standard SGD as the optimizer.

In Table ?? we report the perplexities we obtained with LMN along with the results reported in published work for other three approaches. As the table demonstrates we achieve state of the art results in Text8 and Wikitext2. In LMN the auto-modulation caused by considering only cases where the PCN is weak is superior to memory cache. This shows up in tests when memory is constrained, when the focus on mis-predicted outputs in LMN allows for boosted recall, efficient memory utilization, and capturing longer contexts, compared to other models.

## Conclusion

We extended standard memory models with a label addressable memory module and an adaptive weighting mechanism

| Name         | Wikitext2 (100 ) | Text8 (2000) |
|--------------|------------------|--------------|
| LSTM         | 99.7             | 120.9        |
| Pointer LSTM | 80.8             | -            |
| Neural cache | 81.6             | 99.9         |
| LMN          | <b>77.6</b>      | <b>91.1</b>  |

Table 5: Test perplexity for language modeling on Wikitext and Text8 with memory 100 and 2000 respectively.

for on-line model adaptation. LMNs mix limited data with pre-trained models by combining ideas from boosting and online kernel learning while tapping deep networks to learn representations and RNNs to model the evolving roles of memory and pre-trained models. LMNs have some similarities to recent MANNs but has significant differences. First, we have a label addressable memory instead of content based addressing. Second, we use memory to only store content on which primary network is weak. Third, our model has a loose coupling between memory and network, and hence our model can be used to augment pre-trained models at a very low cost. Fourth we use an adaptive reweighing mechanism to modulate the contribution of memory and PCN. This LMN is demonstrated to be extremely successful on a variety of challenging classification tasks which required fast adaptation to input and handling non-local dependencies. An interesting extension of LMNs is organizing the memory not just based on discrete labels but on learned multi-variate embeddings of labels thereby paving the way for greater sharing among labels.

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